

PROJECT TASK ALLOCATION AND DETAILED SCOPE

Chatbot Development and LLM Research Projects

1. Group Structure

Track	Groups	Assigned Project
Track 1	AI_01, AI_04, AI_07	Inquisitor Chatbot
Track 2	AI_02, AI_05, AI_08	Solar Industry Chatbot
Track 3	AI_03, AI_06, AI_09	MaaProject Chatbot
Track 4	AI_010	LLM Research, Training and Evaluation

Groups within the same track have the same project scope and requirements. Each group should independently develop its solution while following the assigned requirements.

2. Common Requirements for All Chatbot Groups

- Requirement analysis and problem identification.
- Identification of target users and practical chatbot use cases.
- Feature planning before implementation.
- Conversation flow design for major user scenarios.
- Selection and documentation of suitable technologies.
- Development of a functional chatbot prototype.
- Integration with the assigned website or mobile application where applicable.
- Creation and documentation of the chatbot knowledge base.
- Testing with normal, incomplete, invalid, unrelated, and out of scope queries.
- Performance and response quality evaluation.
- Complete technical and user documentation.
- Final presentation and live demonstration.

3. Track 1: Inquisitor Chatbot

Groups: AI_01, AI_04, AI_07

Objective: Develop an intelligent chatbot for the Inquisitor platform that assists users by answering questions, providing information, guiding users, and helping them interact with relevant platform features.

3.1 Requirement Analysis

- Study the Inquisitor platform and its relevant modules.
- Identify target users and their information needs.
- Identify common questions and common user problems.
- Identify available services, features, events, memberships, internships, and support information.
- Define the exact scope and boundaries of the chatbot.

3.2 Core Use Cases

- Frequently asked questions.
- Platform and service information.
- User guidance and navigation assistance.
- Event information.
- Membership information.
- Internship information.
- Registration assistance.
- General inquiries.
- Contact or support assistance.
- Fallback responses when the requested information is unavailable.

3.3 Core Features

- Natural language interaction.
- Knowledge based responses.
- Context aware responses where technically feasible.
- Suggested questions.
- Fallback and error handling.
- Conversation management.
- Relevant links or navigation support.
- Human support or contact option where required.

3.4 Optional Advanced Features

- Conversation history.
- User specific responses.
- Voice input or voice output.
- Multilingual support.
- Personalized recommendations.
- Search functionality.
- Integration with application data.
- User authentication.
- Analytics.

3.5 Technical Documentation

- Frontend technology.
- Backend technology.
- LLM or chatbot technology.
- APIs.
- Database.
- Knowledge base.
- Hosting or deployment.
- Authentication and security.
- Error handling.

3.6 Testing

- Prepare at least 30 test cases.
- Test normal, complex, incomplete, misspelled, unrelated, repeated, out of scope, invalid, long, and multi question queries.

- Record expected response, actual response, result, and required improvements.

3.7 Deliverables

- Working Inquisitor chatbot.
- Source code.
- Requirements document.
- Feature specification.
- Use case document.
- Conversation flow.
- System architecture.
- Knowledge base documentation.
- Testing report.
- Screenshots.
- Installation instructions.
- Technical documentation.
- User guide.
- Presentation.
- Final demonstration.

4. Track 2: Solar Industry Chatbot

Groups: AI_02, AI_05, AI_08

Objective: Develop an intelligent chatbot for a Solar Industry platform that assists customers with solar systems, products, energy consumption, installation, maintenance, and related services.

4.1 Required Industry Research

- Solar panels.
- Solar inverters.
- Batteries.
- On grid systems.
- Off grid systems.
- Hybrid systems.
- Solar system sizing.
- Energy consumption.
- Installation.
- Maintenance.
- Warranty.
- Solar pricing concepts.
- Customer support.

4.2 Core Chatbot Functions

- Product and system information.
- Panel, inverter, and battery information.
- System type explanation.
- Solar capacity and energy consumption questions.
- Installation and maintenance guidance.
- Warranty information.

- Frequently asked questions.
- Customer inquiries.

4.3 Solar Recommendation Feature

- Ask about monthly electricity units.
- Ask about average electricity bill.
- Estimate or ask for daily consumption.
- Ask about available roof area where relevant.
- Ask about required backup time.
- Ask about desired system type.
- Ask about battery requirements.
- Ask about major electrical loads.
- Provide an approximate system recommendation.
- Clearly state that the recommendation is an initial estimate and does not replace a professional site survey or engineering design.

4.4 Optional Advanced Features

- Solar calculator.
- Inverter recommendation.
- Battery recommendation.
- Panel recommendation.
- Estimated generation.
- Approximate savings calculation.
- Customer lead collection.
- Product comparison.
- Installation request.
- Human representative support.

4.5 Technical Requirements

- Chatbot architecture.
- Solar knowledge base.
- LLM or chatbot model.
- Backend.
- Frontend.
- Database.
- APIs.
- Calculation logic.
- Security.
- Deployment.

4.6 Testing

- Prepare at least 30 test cases.
- Test product questions, sizing questions, consumption questions, technical questions, customer service questions, invalid questions, unrelated questions, incomplete information, and multiple questions.

4.7 Deliverables

- Working Solar Industry chatbot.
- Source code.

- Solar knowledge base.
- Requirements document.
- Use case document.
- Conversation flows.
- Solar recommendation logic.
- System architecture.
- Test report.
- Screenshots.
- Technical documentation.
- User guide.
- Installation instructions.
- Presentation.
- Final demonstration.

5. Track 3: MaaProject Chatbot

Groups: AI_03, AI_06, AI_09

Objective: Develop an intelligent chatbot for MaaProject that helps users access information, understand services, find relevant information, and receive assistance with common queries.

5.1 MaaProject Analysis

- Study the project purpose.
- Identify target users.
- Identify available services.
- Identify main platform features.
- Identify common user questions.
- Identify common user problems.
- Identify available information.
- Identify potential chatbot applications.

5.2 Core Functions

- Frequently asked questions.
- Project information.
- Service information.
- User guidance.
- Navigation assistance.
- Search assistance.
- General inquiries.
- Contact information.
- Problem reporting.
- Support requests.

5.3 Optional Advanced Features

- Conversation history.
- Personalized responses.
- User authentication.
- Search integration.

- Database integration.
- Document based question answering.
- Multilingual support.
- Voice interaction.
- Human support escalation.
- Analytics.

5.4 Knowledge Base

- MaaProject website information.
- Project documentation.
- FAQs.
- Service descriptions.
- User guides.
- Approved project documents.
- Database information where applicable.

5.5 Conversation Design

- Create at least 10 complete conversation flows covering common user scenarios.

5.6 Testing

- Prepare at least 30 test cases.
- Test normal, complex, incomplete, incorrect, unrelated, repeated, out of scope, invalid, long, and multiple question queries.

5.7 Deliverables

- Working MaaProject chatbot.
- Source code.
- Requirements document.
- Knowledge base.
- Feature specification.
- Use cases.
- Conversation flows.
- System architecture.
- Test report.
- Screenshots.
- Technical documentation.
- User guide.
- Installation instructions.
- Presentation.
- Final demonstration.

6. Track 4: LLM Research, Training and Evaluation

Group: AI_010

Objective: Research, train or fine tune where appropriate, and evaluate Large Language Models for different industries. Identify practical free and open source solutions that can be used for chatbot development without mandatory licensing costs.

6.1 Research Objectives

- Existing LLMs.
- Open source LLMs.
- Small language models.
- Locally deployable models.
- Cloud based free options.
- Industry specific models.
- RAG based systems.
- Fine tuning approaches.
- Model evaluation methods.
- Deployment strategies.

6.2 Industry Research

- Select at least 5 industries.
- Possible industries include education, healthcare, solar and energy, banking, e commerce, manufacturing, real estate, tourism, customer support, and engineering.
- For each industry document major problems, target users, chatbot applications, required knowledge, data requirements, security, privacy, suitable architecture, benefits, and limitations.

6.3 Model Comparison

- Select at least 3 LLMs for practical comparison.
- Compare model size, parameters, context length, hardware requirements, response quality, speed, accuracy, hallucination, fine tuning support, RAG support, local deployment, license, commercial usage, and estimated cost.

Criteria	Model 1	Model 2	Model 3
Model size			
Parameters			
Context length			
Hardware requirements			
Response quality			
Response speed			
Accuracy			
Hallucination			
Fine tuning support			
RAG support			
Local deployment			
License			
Commercial usage			
Estimated cost			

6.4 Training and Fine Tuning

- Dataset preparation.
- Data cleaning.
- Prompt design.
- Model testing.
- Fine tuning or parameter efficient fine tuning.
- Evaluation.
- Result comparison.
- If full model training is computationally impractical, document the reason and perform a smaller fine tuning or adaptation experiment.

6.5 RAG Research

- Study Retrieval Augmented Generation.
- Develop a small RAG prototype.
- Demonstrate the flow: user question, document retrieval, relevant information, LLM, final response.

6.6 Evaluation

- Accuracy.
- Relevance.
- Response quality.
- Hallucination.
- Response time.
- Resource consumption.
- Cost.
- Ease of deployment.
- Privacy.
- Scalability.

6.7 Final Recommendations

- Best model for low resource computers.
- Best model for local deployment.
- Best model for customer support.
- Best option for RAG chatbots.
- Best option for industry specific applications.
- Best option for mobile applications.
- Best option for website chatbots.
- Best option for research and experimentation.

6.8 Deliverables

- Research report.
- Industry analysis.
- LLM comparison.
- Dataset.
- Training or fine tuning experiment.
- RAG prototype.
- Evaluation framework.
- Experimental results.
- License comparison.
- Cost comparison.
- Final recommendations.
- Source code.
- Presentation.
- Final demonstration.

7. Common Evaluation Criteria

Category	Weight
Requirement analysis	10%
Research and use cases	10%

Feature implementation	20%
Chatbot accuracy and quality	15%
UI and user experience	10%
Technical architecture	10%
Testing and evaluation	10%
Documentation	5%
Presentation	5%
Final demonstration	5%
Total	100%

8. Important Instructions

1. AI_01, AI_04, and AI_07 have the same Inquisitor chatbot scope.
2. AI_02, AI_05, and AI_08 have the same Solar Industry chatbot scope.
3. AI_03, AI_06, and AI_09 have the same MaaProject chatbot scope.
4. AI_010 has a separate LLM research and evaluation project.
5. Every group must independently develop and document its assigned project.
6. Every group must maintain a clear record of development progress.
7. Every chatbot must have documented test cases and test results.
8. Groups should use real project information rather than unsupported or invented features.
9. Any external model, API, dataset, library, or framework must be properly documented.
10. Open source and free software licenses must be checked before use.
11. Final projects must be functional prototypes rather than interface designs only.
12. Each group must submit both the working project and complete documentation.
13. Each group must be able to explain its implementation, architecture, chatbot logic, testing methodology, and limitations during final evaluation.

9. Final Submission Checklist

- Working project.
- Complete source code.
- Requirements document.
- Feature specification.
- Use case document.
- Conversation flows.
- System architecture.
- Knowledge base documentation.
- Testing report with at least 30 test cases for chatbot groups.
- Screenshots and evidence of implementation.
- Installation and setup instructions.
- Technical documentation.
- User guide.
- Presentation slides.
- Final live demonstration.